Evaluating County Forecasts in the California Omicron Wave via the Allocation Score

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Thank you to my coauthors...

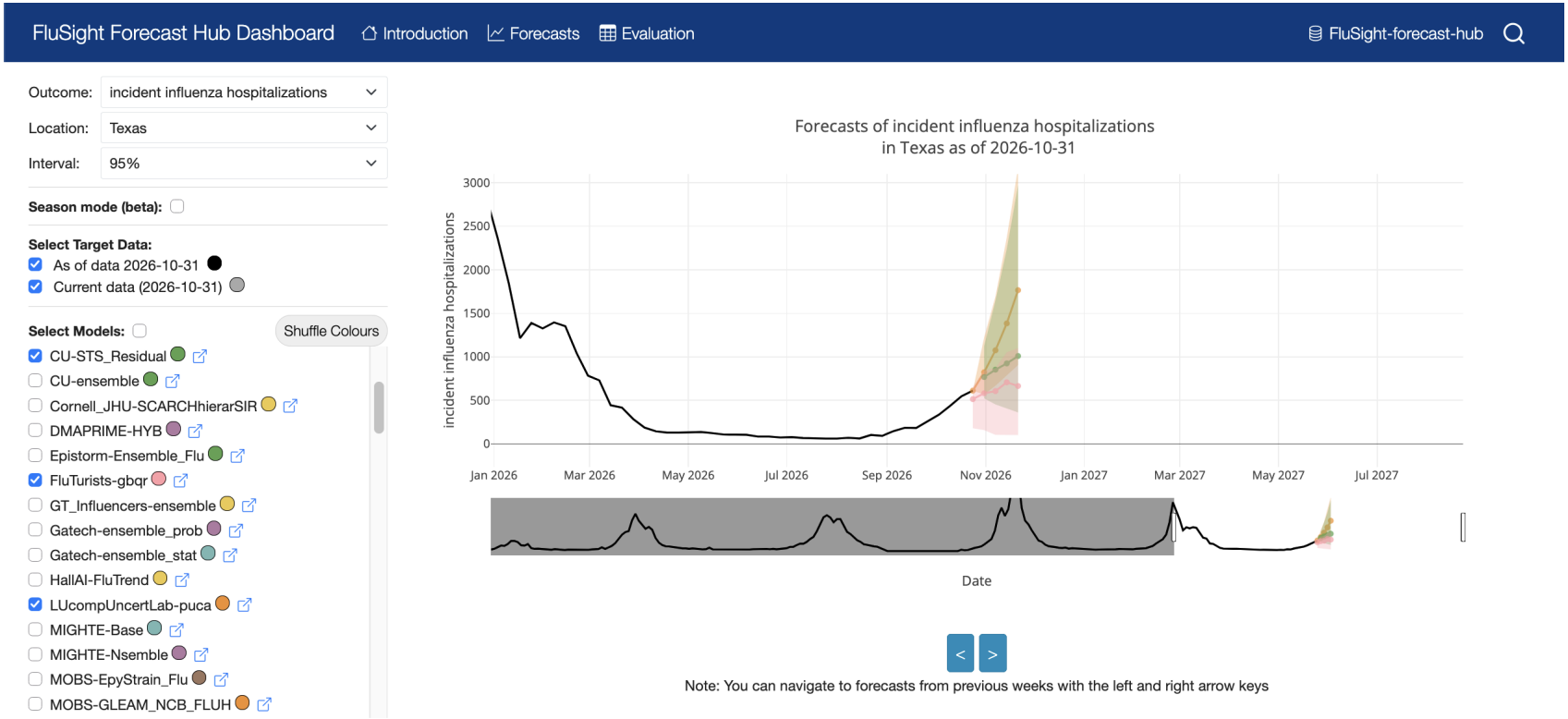
- **Nicholas Reich, PhD** (University of Massachusetts, Amherst)
- **Lauren White, PhD** (California Department of Public Health)
- **Tomás León, PhD** (California Department of Public Health)

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Background

What do we use forecasts for?



Possible Uses For Infectious Disease Forecasting

- Public-facing communications
- Situational Awareness
- Coordination of resources (ex. vaccines, medicines, hospital staffing)
- Advance warning for organizational/emergency activation

The Need for Use-Aware Forecast Evaluation

- Productive engagement between forecasters and practitioners requires that models be created, evaluated, and improved upon with both the public health need and the priorities of the public health agency in mind.
- Forecast evaluation methods are always grounded in assumptions of what constitutes a "best" model.
- Key in interpreting forecast scores:
  - *What does this scoring method assume?*
  - *What does it prioritize?*
  - *Therefore, what is this score saying?*

The Need for Use-Aware Forecast Evaluation

Priority	Metric(s)	Use-Case
Point estimate is close to the truth	MAE, RMSE	Public-facing communications
Forecast interval contains the truth	Coverage, WIS	Situational awareness
Forecast interval is narrow	WIS	Coordination of hospital staffing
Sets of forecasts rank correctly relative to each other	Variogram Score, Allocation Score	Vaccination campaign prioritization
Forecast effectively guides resource allocation decisions across multiple locations	Allocation Score	Allocation of ventilators to hospitals
Forecast correctly predicts rises/falls/leveling	Brier Score	Disease "weather reports"
Forecast correctly predicts which "activity level" the week will fall in	Brier Score	Advance warning for emergency activation

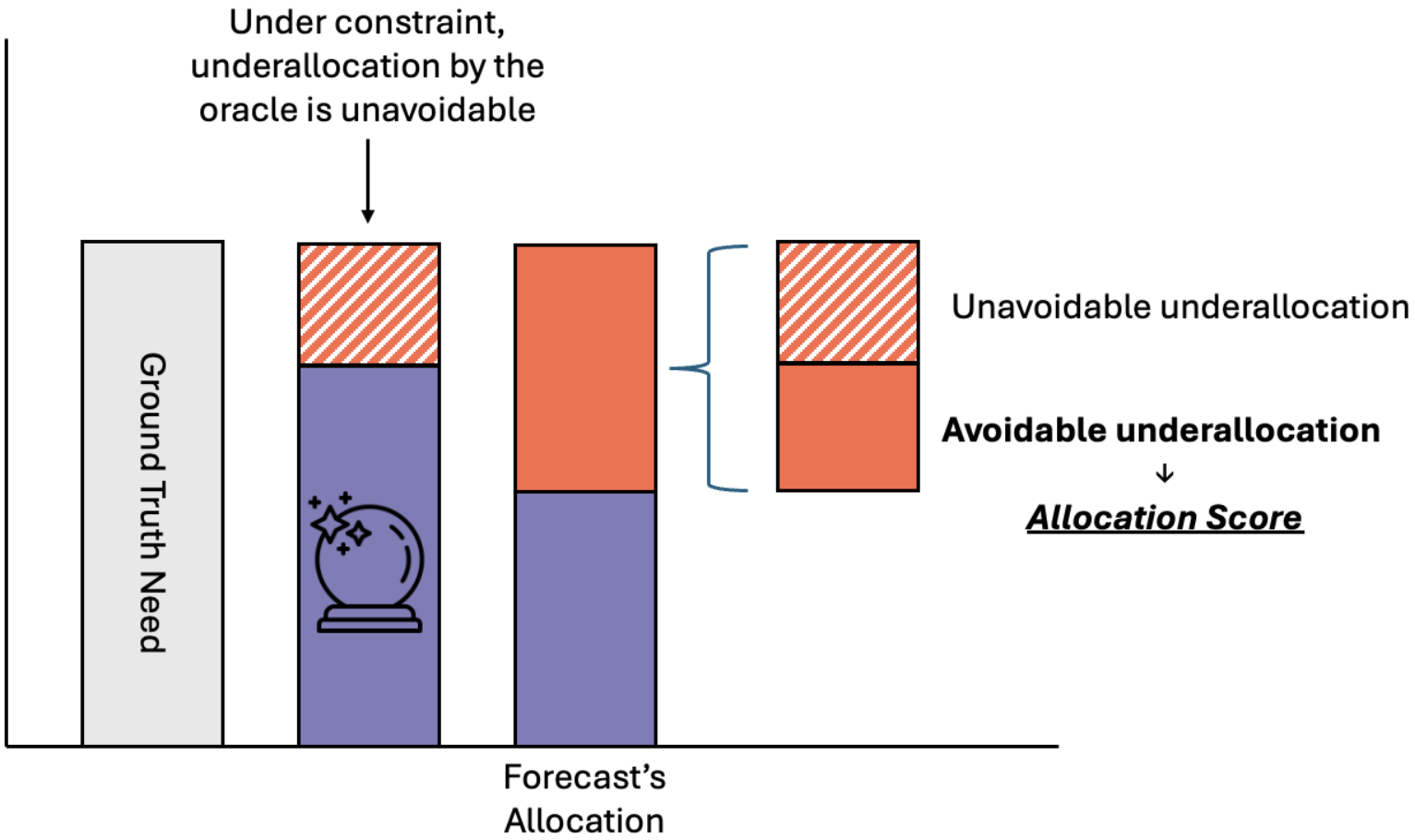
Overview of Scoring Methodologies for Probabilistic Forecasts

- With these priorities in mind, we will:
  - Introduce a recently-developed scoring rule, the Allocation Score (AS)
  - Introduce its extension, the Integrated Allocation Score (IAS)
  - Review the Weighted Interval Score (WIS), another commonly-used evaluation method for probabilistic forecasts.
- These 3 methods will be the focus of this analysis.

Allocation Score (AS)

- Original paper: *Evaluating infectious disease forecasts with allocation scoring rules* (Gerding, Reich, Rogers, & Ray, 2024)
- Treats forecasts as predicting **resource need** at the location of interest. Based on this, scores forecasts on their success in **guiding allocation of limited resources across locations**.
- "**Oracle**" scheme: an ideal resource allocation, done with knowledge of the future.
- Compares the underallocation of the forecast to that of the oracle scheme to determine avoidable underallocation.
- **Lower score = less avoidable unaddressed need = better forecast**

Allocation Score (AS)



Allocation Score Procedure

K=20,000	Allocation by population	Allocation by Forecast 1	Allocation by Forecast 2	True resource need
Rural county Pop 100,000	1,000	100	400	700
Suburban county 1 Pop 400,000	4,000	4,200	1,000	5,000
Suburban county 2 Pop 400,000	4,000	2,500	300	2,100
Urban county Pop 1,100,000	11,000	13,200	18,300	13,000

Total = 20,800
Unavoidable loss = 800

Allocation Score Procedure

K=20,000		Allocation by population	Allocation by Forecast 1	Allocation by Forecast 2	True resource need
Rural county Pop 100,000		1,000	100 -600	400 -300	700
Suburban county 1 Pop 400,000		4,000 -1,000	4,200 -800	1,000 -4,000	5,000
Suburban county 2 Pop 400,000		4,000	2,500	300 -1,800	2,100
Urban county Pop 1,100,000		11,000 -2,000	13,200	18,300	13,000
		-3,000	-1,400	-6,100	Total = 20,800 Unavoidable loss = 800

Allocation Score Procedure



K=20,000



Allocation by population



Allocation by Forecast 1



Allocation by Forecast 2



True resource need

	3,000	1,400	6,100	Total = 20,800
Allocation score =	2,200	600	5,300	Unavoidable loss = 800

Allocation Score Procedure

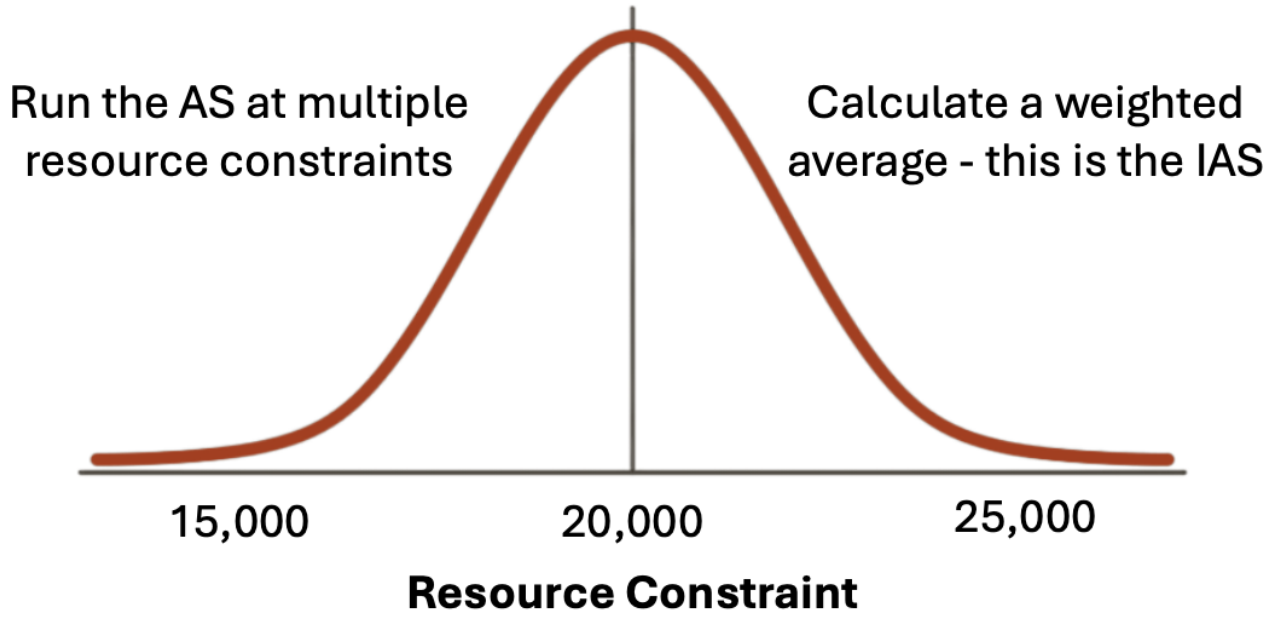


Allocation Score (AS)

- While the resource in question remains abstract, this may be more meaningful to decision-makers in some public health contexts.
- Produces a measurement of avoidable, unaddressed resource need in a scenario where the forecast directly informs allocation decisions.
- Rewards forecasts that correctly capture relative resource needs across locations.

Integrated Allocation Score (IAS)

- Addresses uncertainty with regards to resource constraints
- AS is calculated across a series of possible constraints and a weighted average is produced based on their perceived probabilities.



AS and IAS Considerations

- Score probabilistic forecasts.
- Measure underallocation, not overallocation.
  - While we might see different costs for under-/overallocation in practice, the AS framework assumes no cost to overallocating.
- Measure underallocation in the absolute sense
  - Locations with large populations/needs will contribute more to the score.
- Relative magnitudes of forecast predictions are more important than the forecasted values themselves.

Weighted Interval Score (WIS)

- Approximates the continuous ranked probability score (CRPS) for probabilistic forecasts expressed as prediction intervals.
- Lower score = better
- Contains terms for the weighted sum of the probability interval widths and a penalty for any intervals not including the eventual observation.
  - Therefore, the best-scoring forecasts for WIS are those that are both sharp and well-calibrated to the target of interest.
- WIS evaluates each forecast for a singular horizon, date, and target
- Mean WIS (MWIS) is used to assess and compare forecasts over time and for various forecast targets.

Forecasting Hub and California Forecasting Data

- This analysis focuses on county-level, incident COVID-19 cases in the State of California during the omicron epidemic.
- Data were sourced from the COVID-19 Forecast Hub, which serves as a central repository of forecasts and predictions submitted by international research groups.
- Submissions are aggregated in a standardized format, enabling visualization, scoring, and analysis.

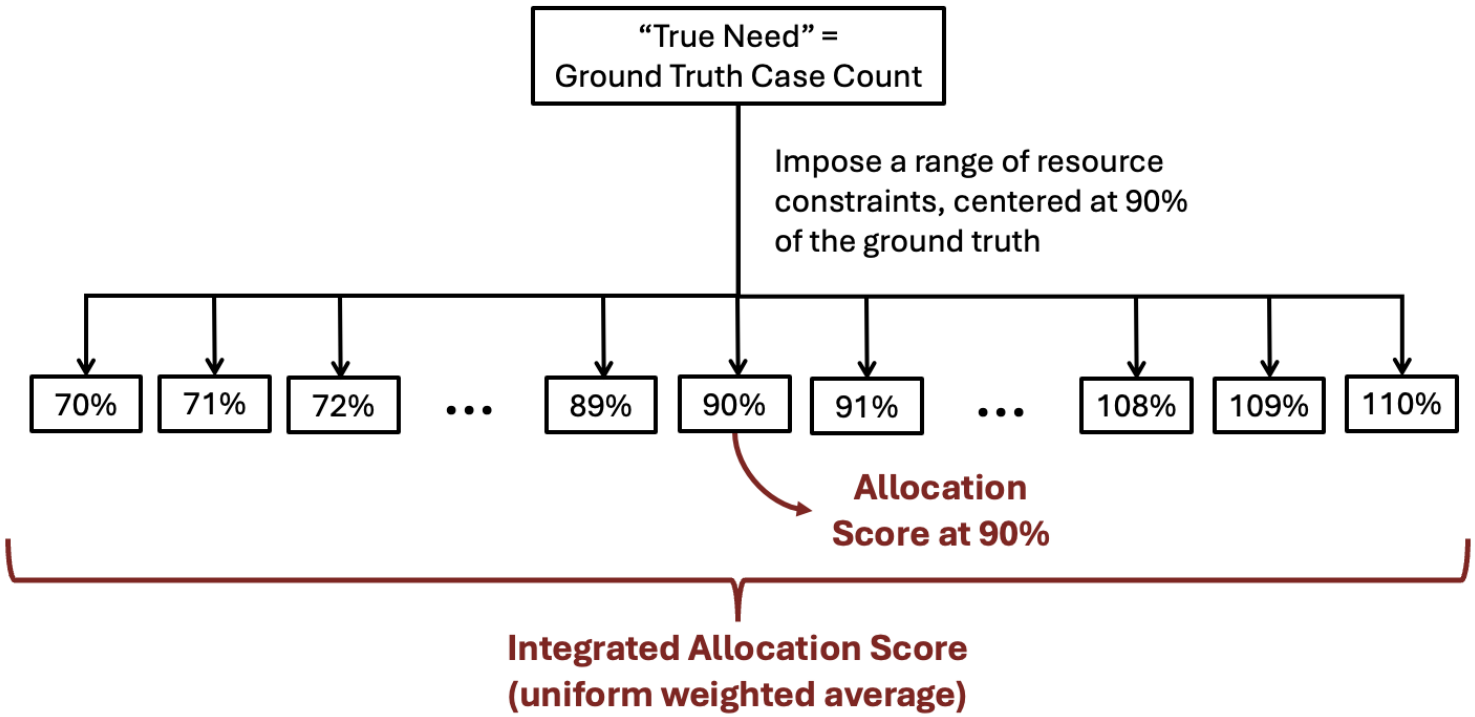
Experimental Design

Overview

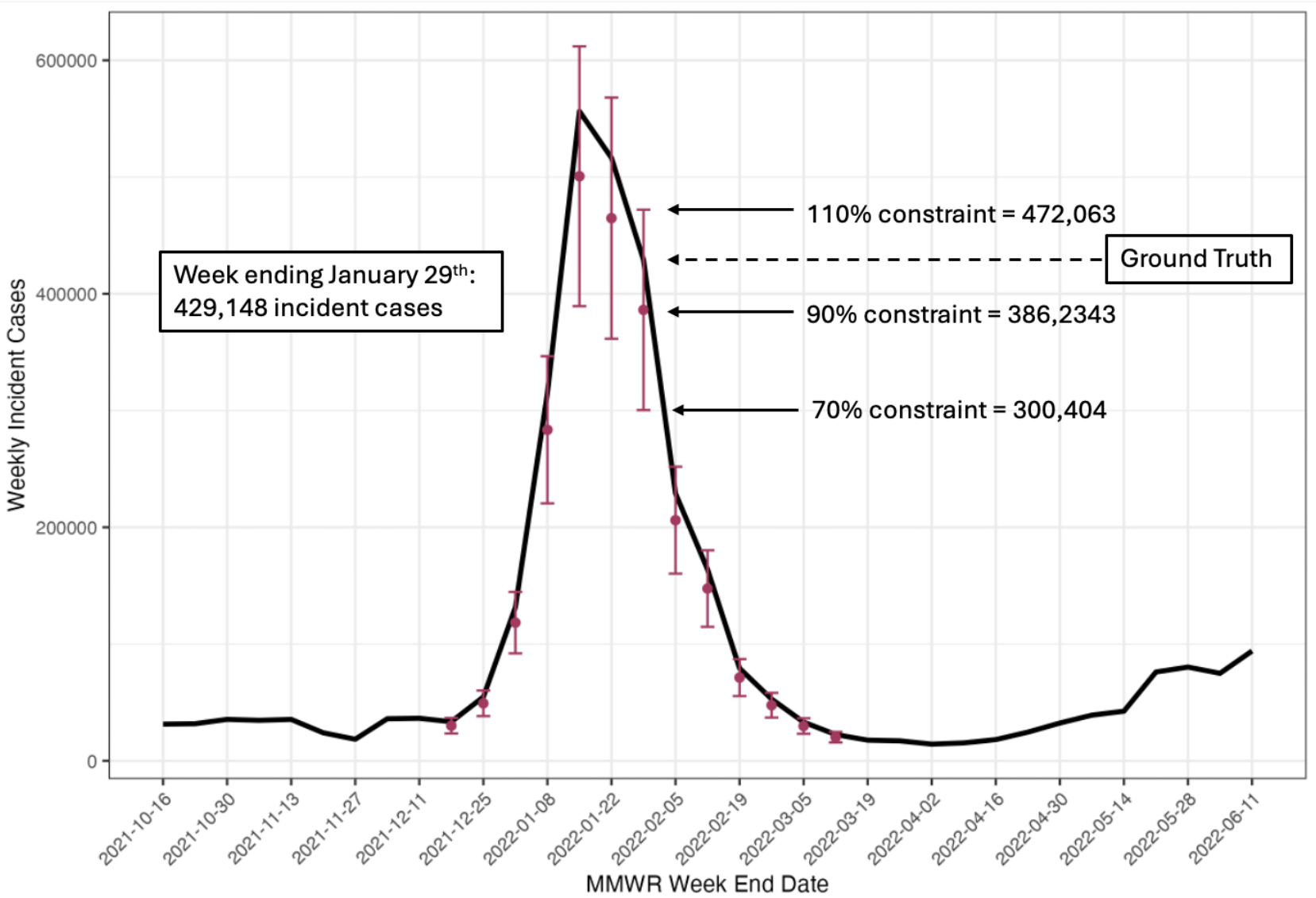
- Forecasts from the COVID-19 Forecast Hub were extracted for13 target dates covering the the omicron wave (12/18/2021 to 03/12/2022), at both the 1 and 3 week horizons.
- Target: county-level weekly incident COVID-19 case counts.
- Examined all California counties with the exception of Los Angeles County (n=57)
  - (LA County was dropped due to high population and use of resource distribution lines from the federal government, in contrast to the state level lines used by other counties.)

Overview

- AS was calculated with constraints relative to the ground truth count at each date: one allocation score for each percentage point between 70% and 110% (centered at 90% of true need), using the `alloscore` package.
- IAS was calculated as a uniform-weighted average (mean) of the AS values.



Overview


600000
400000
200000
0
Weekly Incident Cases
Week ending January 29th:
429,148 incident cases
110% constraint = 472,063
Ground Truth
90% constraint = 386,2343
70% constraint = 300,404
2021-10-16
2021-10-30
2021-11-13
2021-11-27
2021-12-11
2021-12-25
2022-01-08
2022-01-22
2022-02-05
2022-02-19
2022-03-05
2022-03-19
2022-04-02
2022-04-16
2022-04-30
2022-05-14
2022-05-28
2022-06-11
MMWR Week End Date

Overview

- MWIS was calculated as the mean WIS across all 57 counties for a given target date and horizon, using the `scoringutils` package.
- AS at the 90% constraint, IAS, and MWIS were compared.
- Will the scores agree on the "best" model?

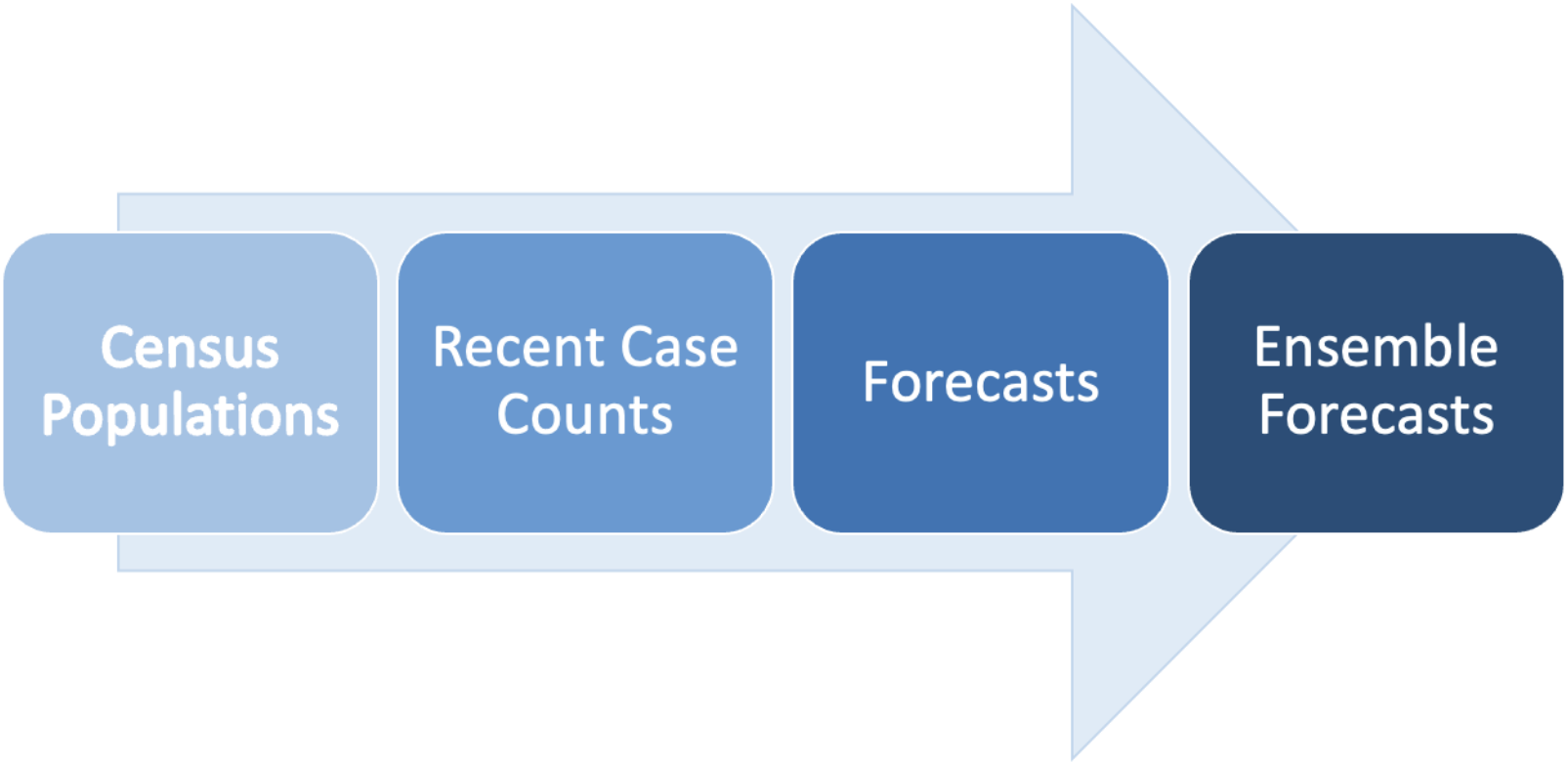
Forecast Selection

- Eligible forecast submissions were defined as submissions for the target dates of interest, at the 1 or 3 week horizon, and providing the 7 required quantiles for all 57 counties of interest.
- Models submitting for less than half of the target dates were also dropped.
- All remaining predictions for a given target and horizon were incorporated into the **eligible-models-ensemble** with the exception of the COVIDhub-4_week_ensemble. The ensemble was produced by taking the simple mean of predictions at each quantile.

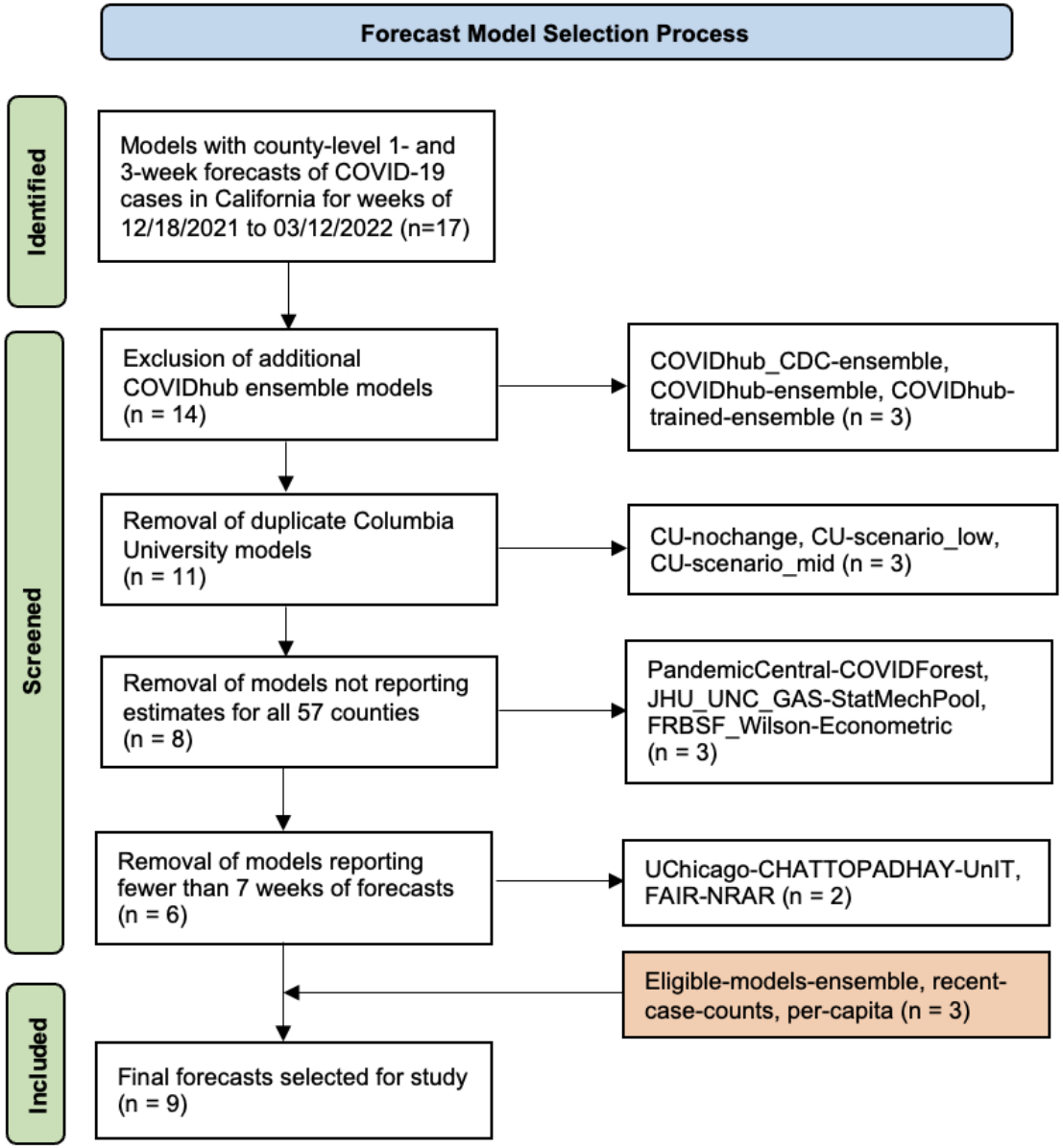
Additional Allocation Schemes Evaluated

- In addition to forecasts, 2 controls were evaluated.
- **Recent-case-counts**: resources are allocated proportional to the most recently reported case count
  - *Equivalent to public health decision-making based on case reporting*
- **Per-capita**: resources are allocated proportional to census population
  - *Equivalent to public health decision-making based on population*

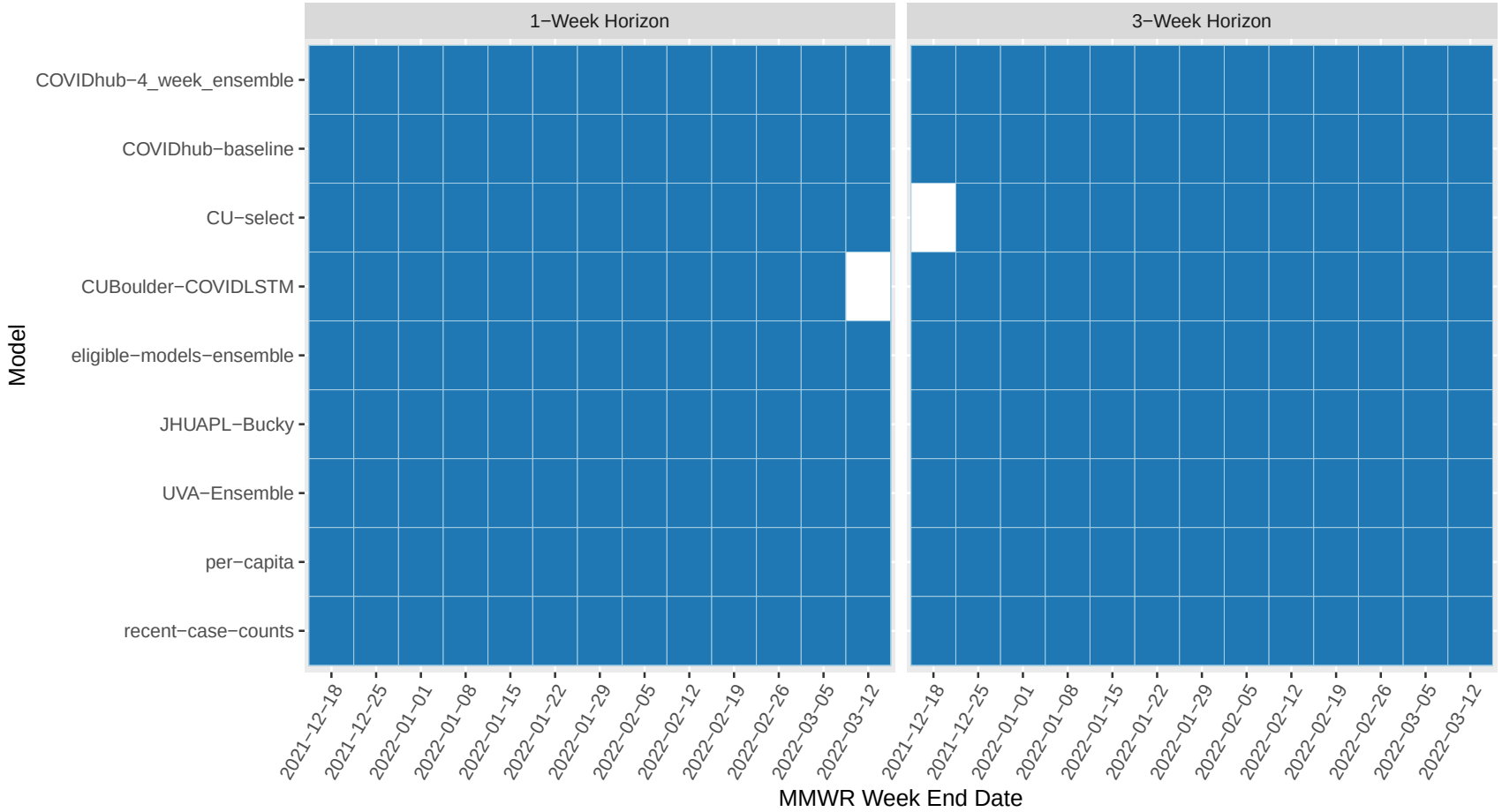
Hierarchy of Information



Forecasts and Allocation Schemes Evaluated

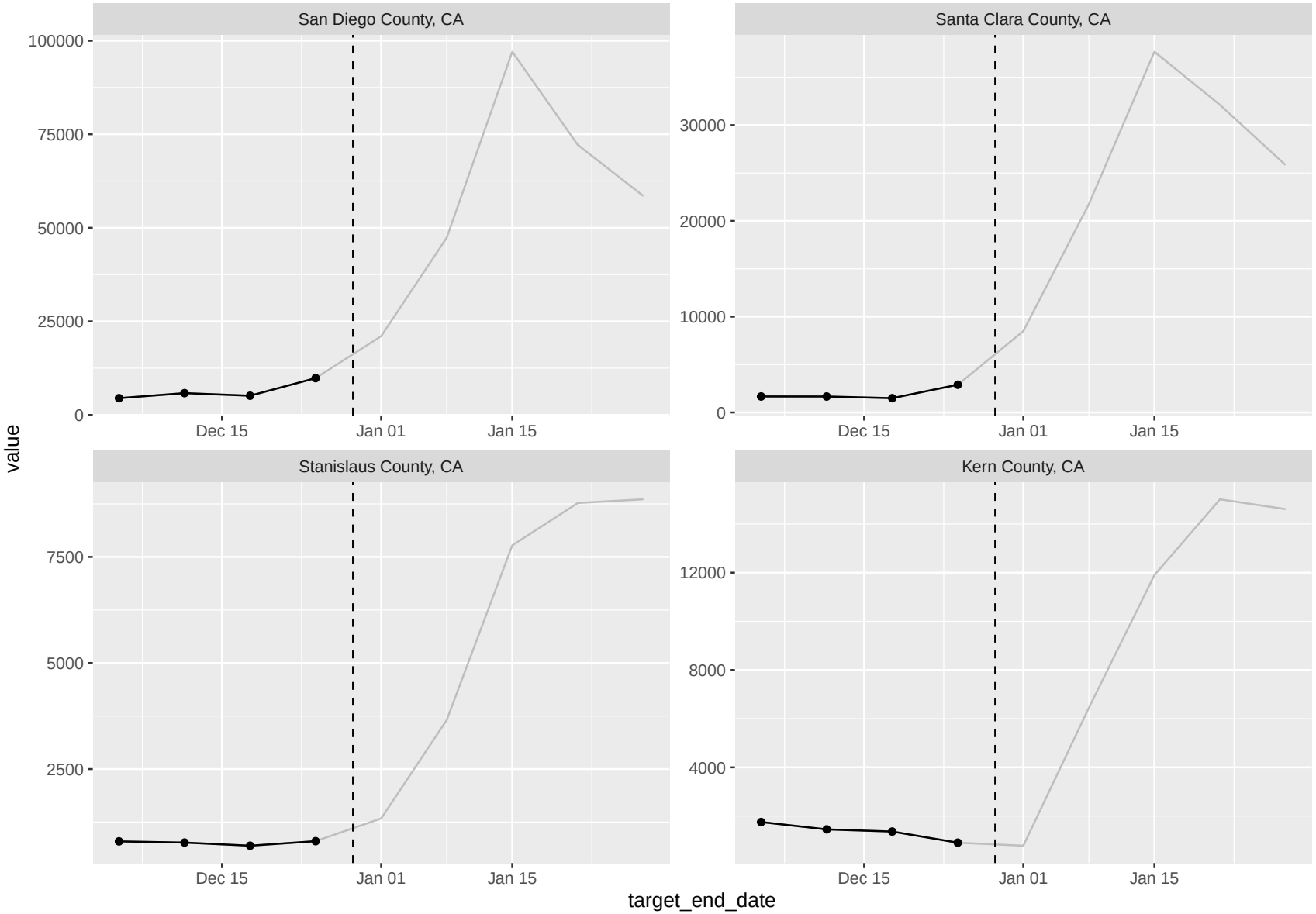


Reporting Completeness

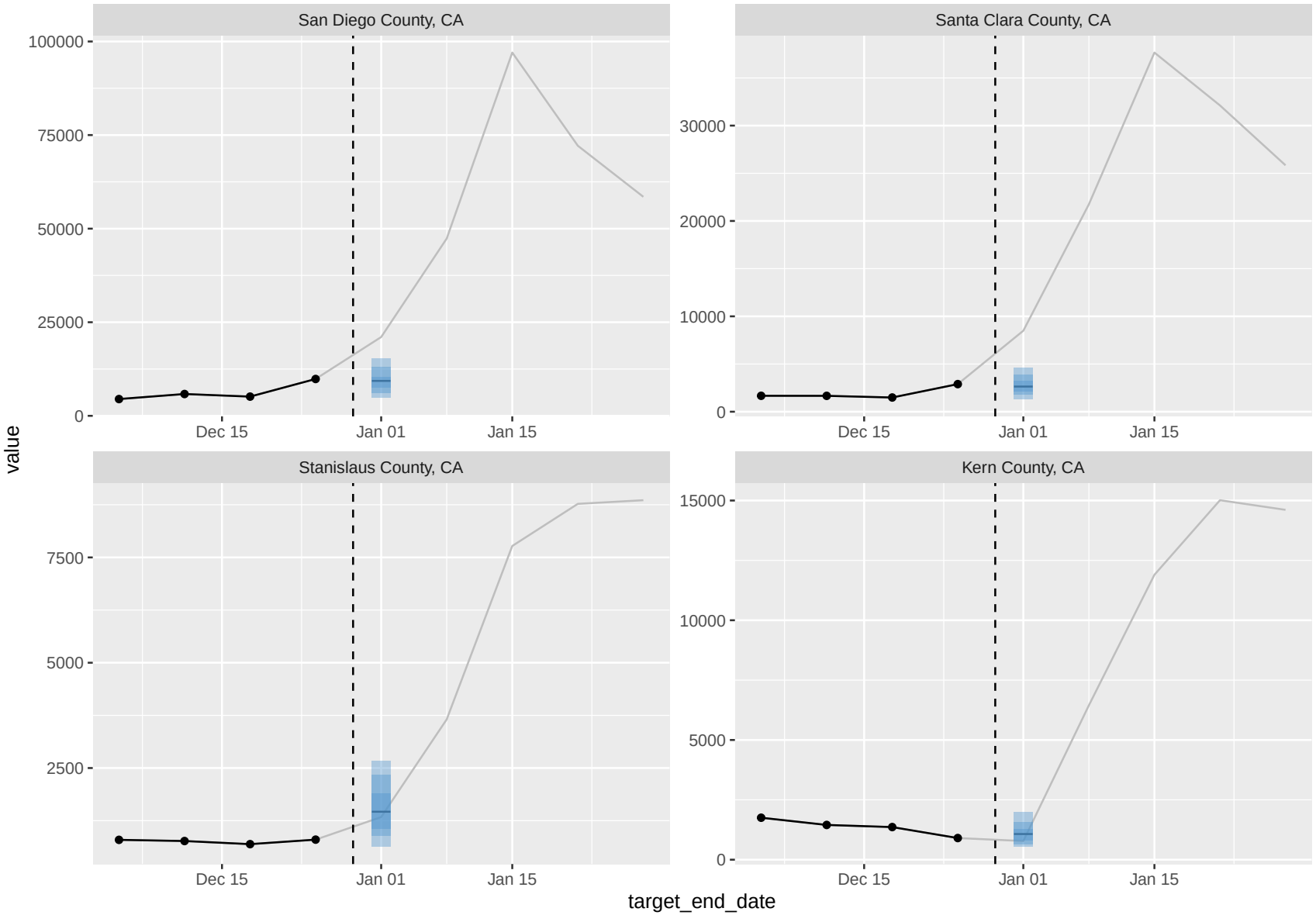


Results

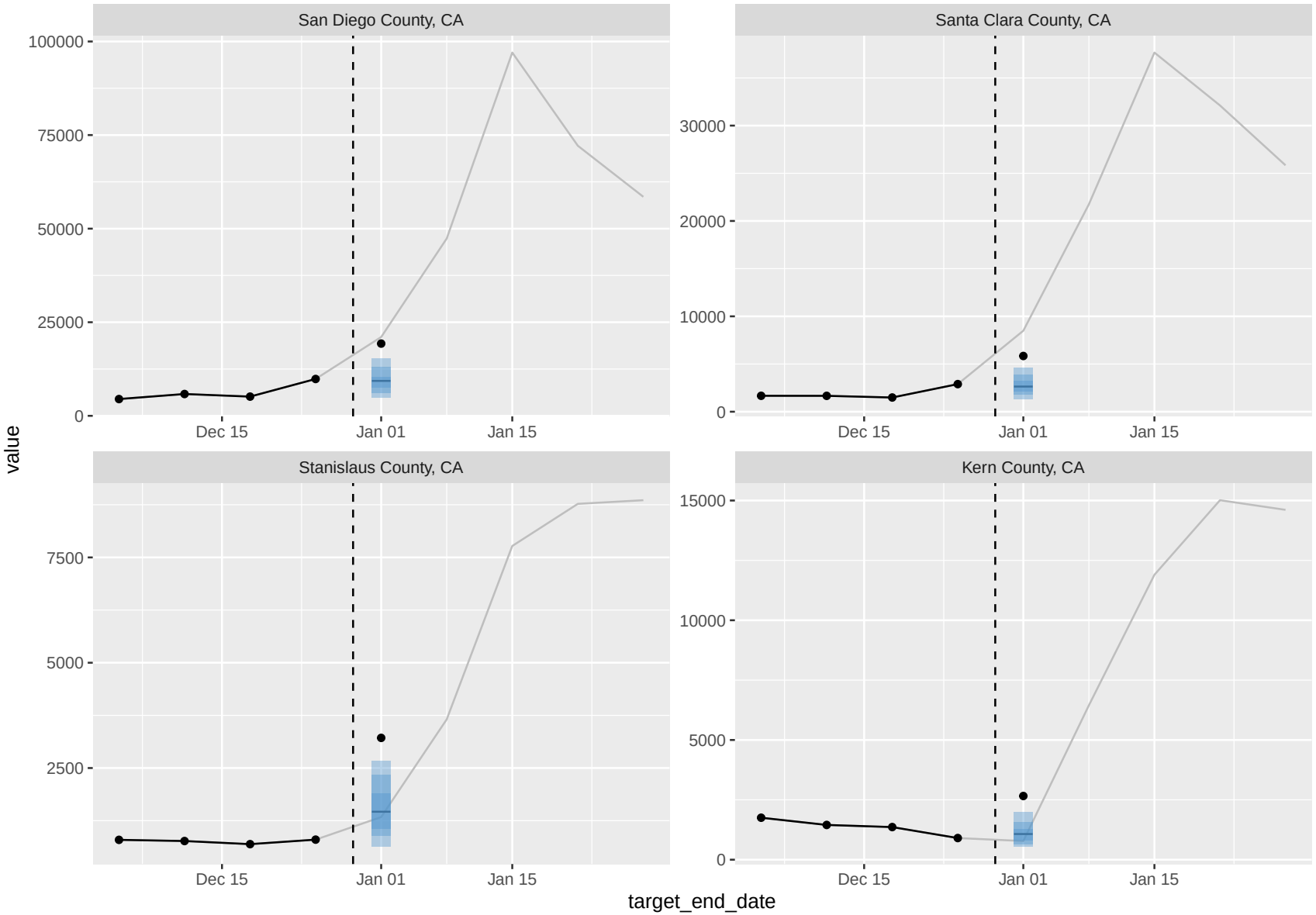
Allocation Example: COVIDhub-4_week_ensemble, 01-01-2022



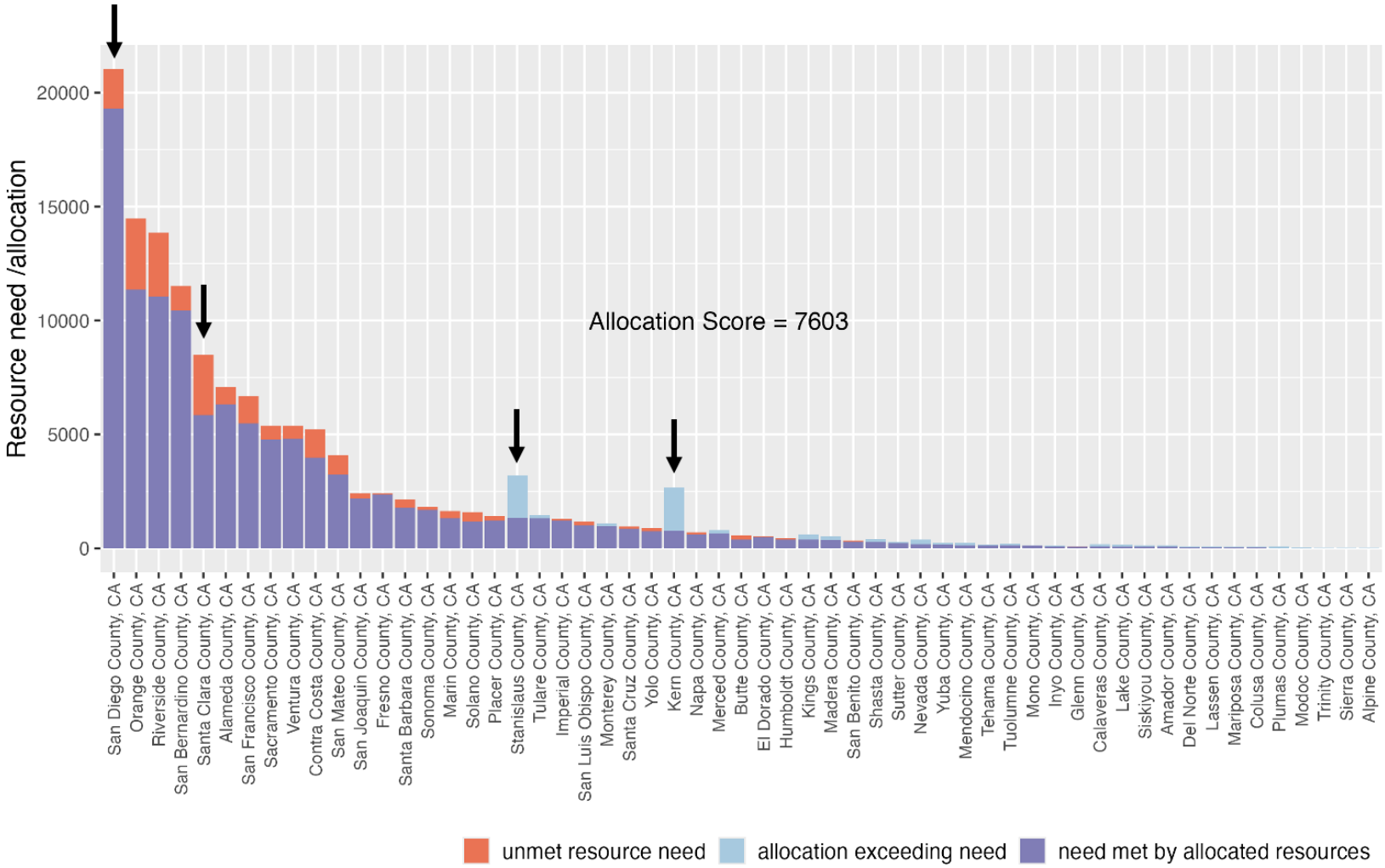
Allocation Example: COVIDhub-4_week_ensemble, 01-01-2022



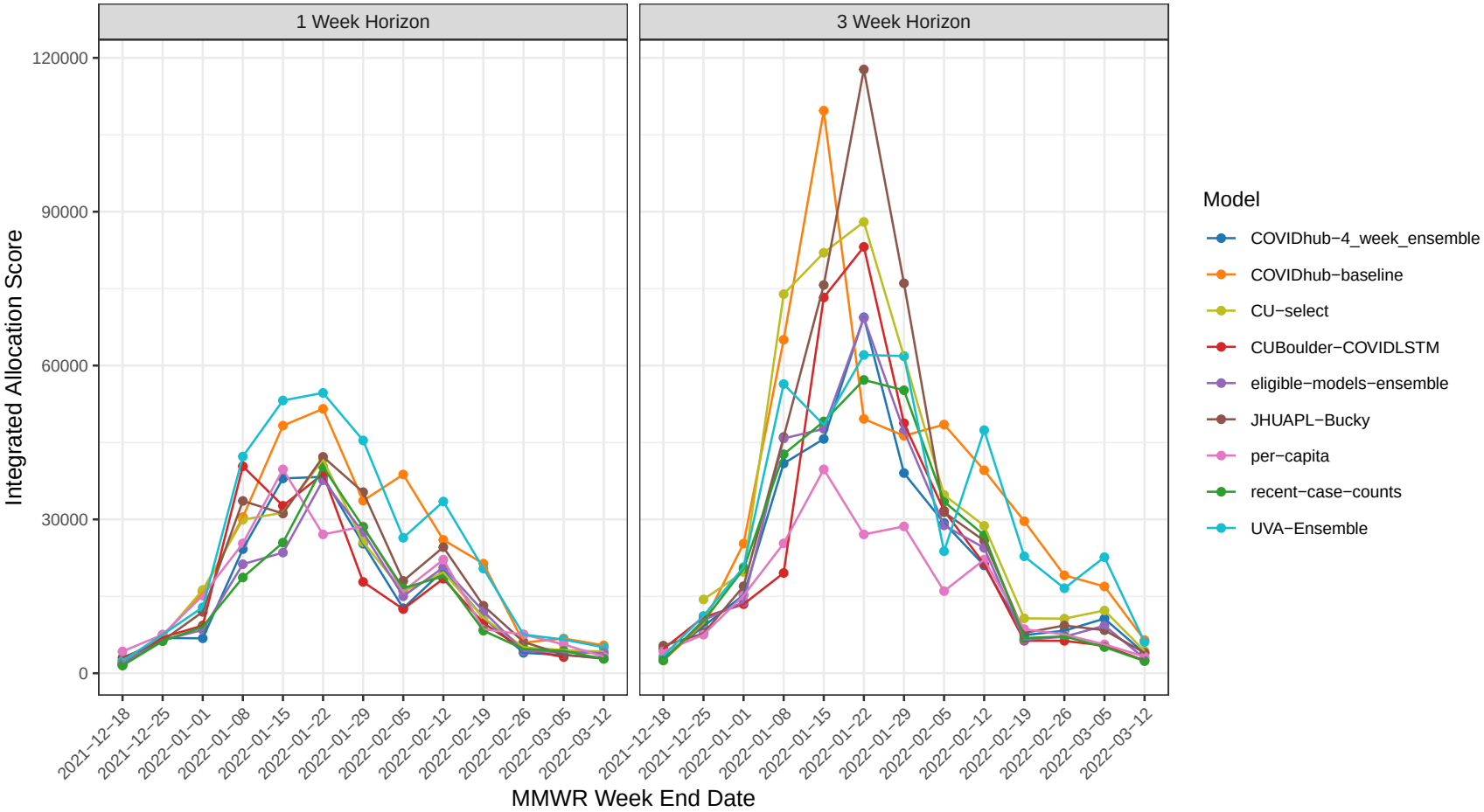
Allocation Example: COVIDhub-4_week_ensemble, 01-01-2022



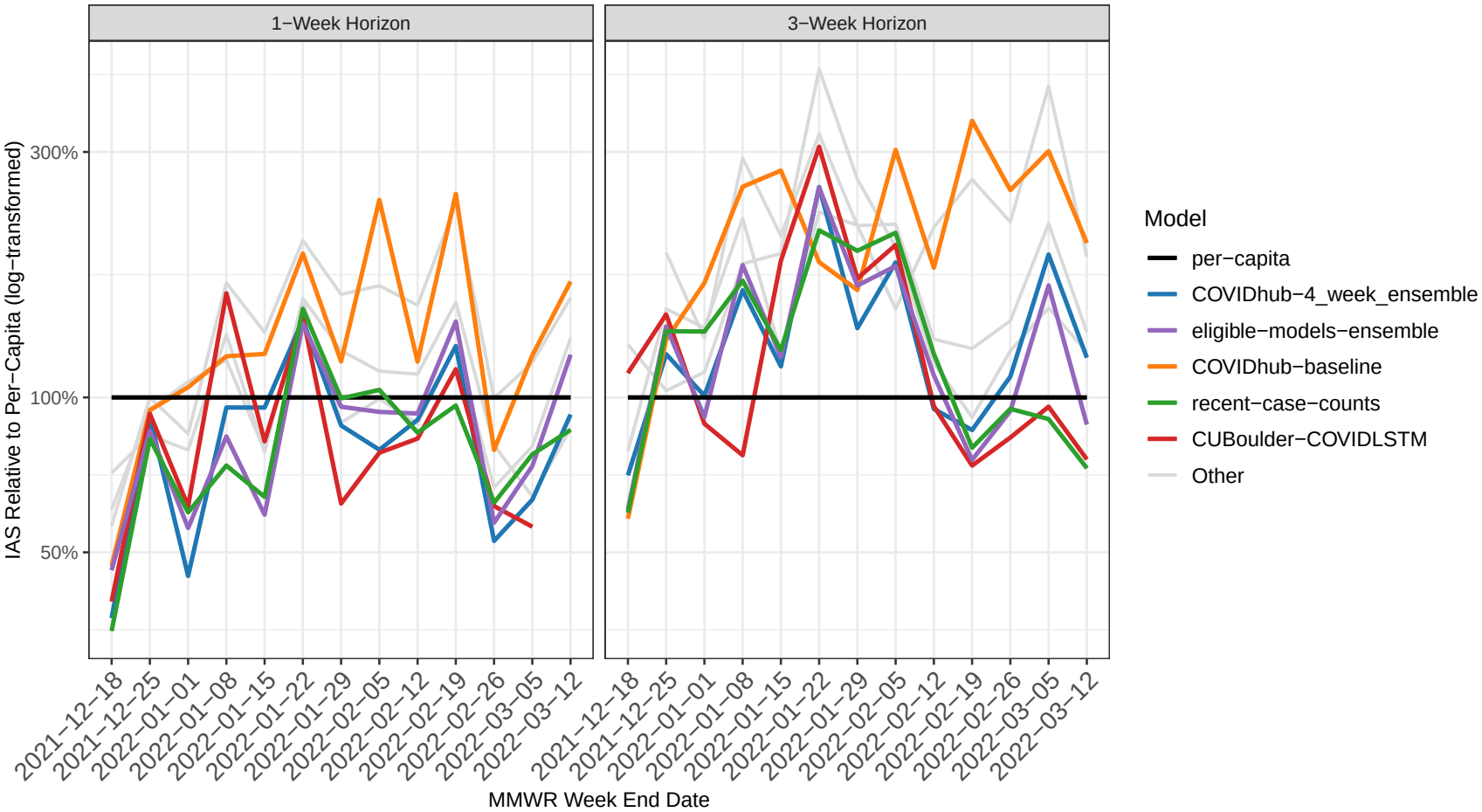
Allocation Example: COVIDhub-4_week_ensemble, 01-01-2022



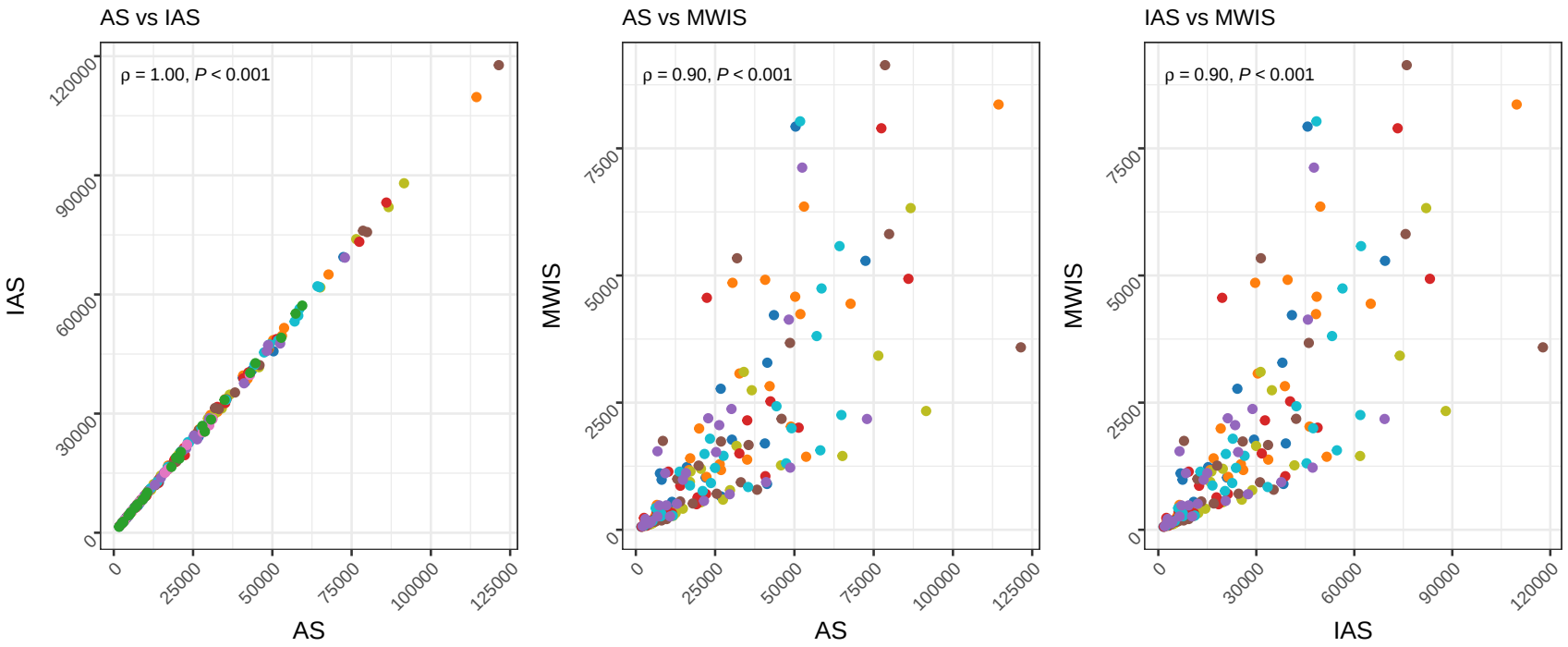
Scores scaled with the epidemic curve and were higher at the 3 week horizon.



Forecasts often scored worse than the per-capita control and even some of the better performers struggled to outperform recent-case-counts.

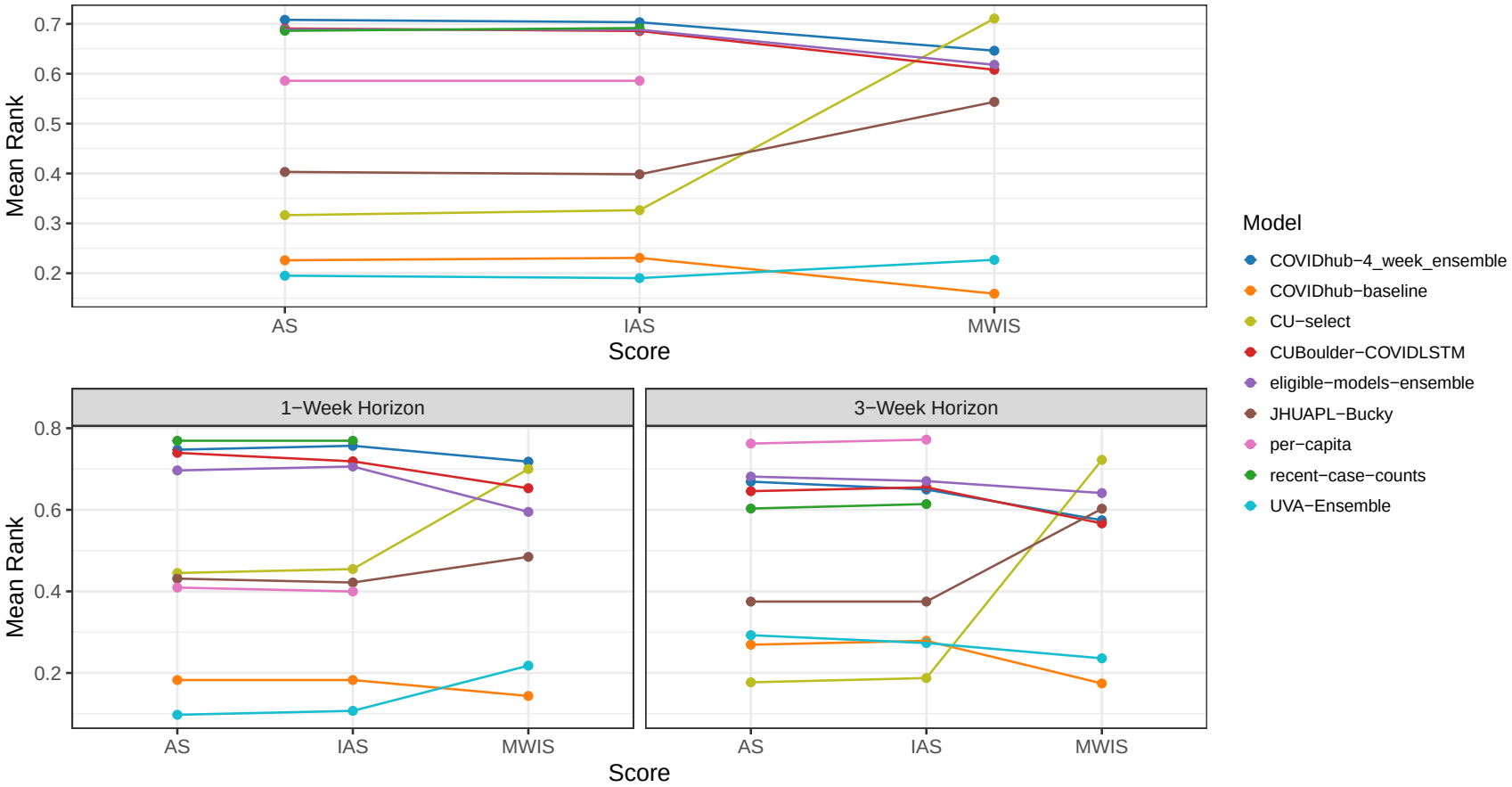


High scores in IAS were nearly perfectly correlated with high scores in AS. MWIS was also very closely correlated with IAS and AS.



Note: per−capita and recent−case−counts weren't scored by MWIS, as they don't provide a predictive distribution.

AS and IAS ranked models the same. MWIS sometimes disagreed.



Note: per−capita and recent−case−counts weren't scored by MWIS, as they don't provide a predictive distribution.

Results Summary

- Scores scaled approximately with the epidemic curve, with the highest scores for predictions of the wave's peak weeks.
- Scores at the 3-week forecast horizon were noticeably higher than those at the 1-week horizon.
- Hub ensemble forecasts (COVIDhub-4_week_ensemble, eligible-models-ensemble) were some of the highest ranking models in all scores and at both horizons. COVIDhub-baseline ranked among the lowest.

Results Summary

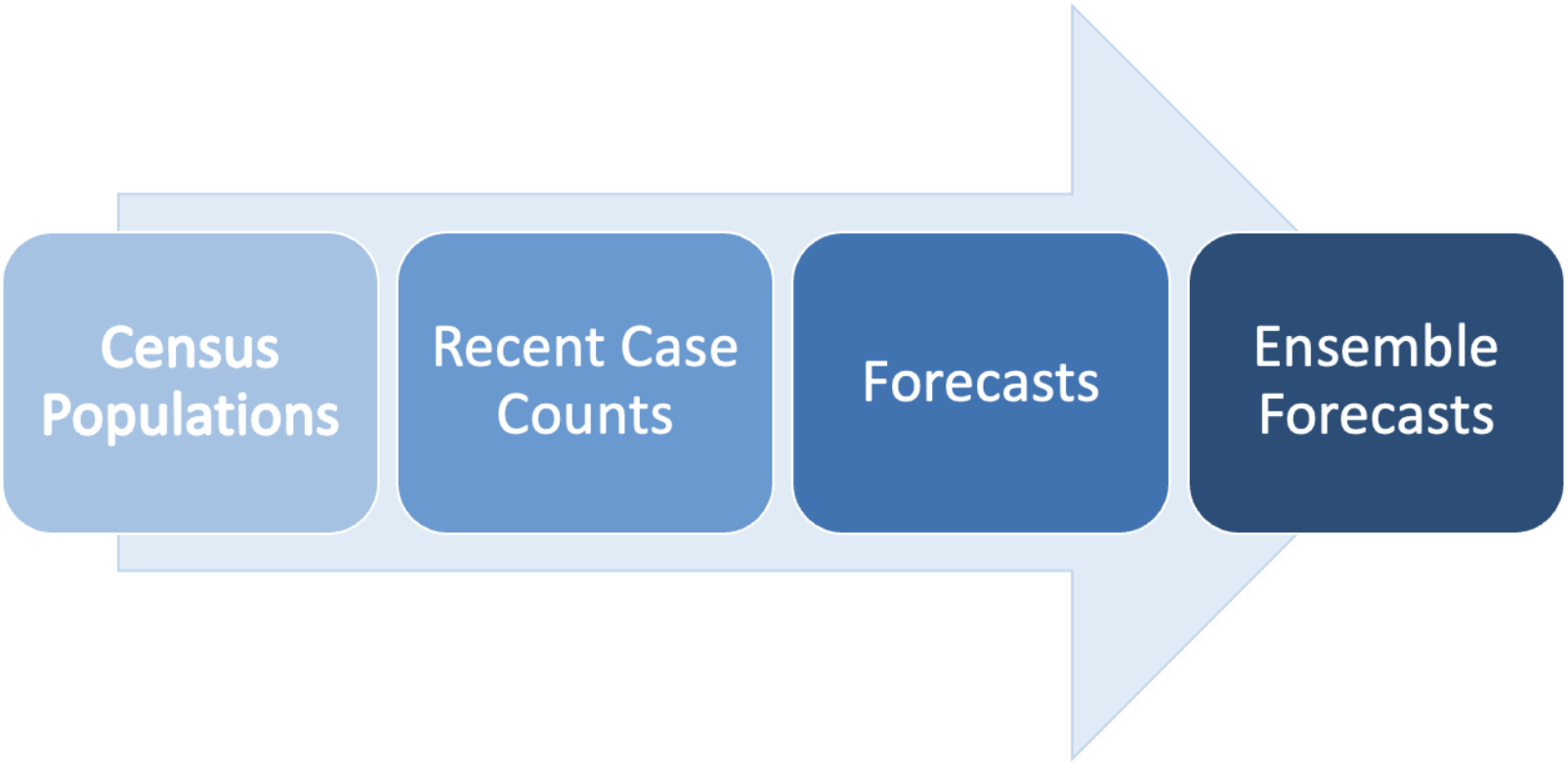
- Forecasts frequently failed to outperform either the per-capita or recent-case-counts control in IAS, particularly at the 3-week horizon.
- The worst relative performance (compared to per-capita) for many models was at the peak of epidemic, when public health decisions can be considered most critical.
- 90% AS and IAS were tightly correlated. High-ranking models in IAS/AS generally ranked highly in MWIS, but there were a few notable exceptions.

Conclusions

Conclusions

- In this analysis, we demonstrated the use of the integrated allocation score to score forecasts of county-level incident COVID-19 cases during California's omicron wave.
- This method measures avoidable, unaddressed resource need in a scenario where the forecast determines policy, rewarding models that capture relative needs correctly across locations.
- We found that many of the county-level forecasts would not have improved allocation decisions during the omicron wave compared to the control methods.
  - **Exceptions:** the Hub ensembles and CUBoulder-COVIDLSTM often scored similarly to/better than per-capita (though not better than recent-case-counts.)

Conclusions



Conclusions

- Differences in model rankings by AS/IAS and MWIS highlight how different priorities and measurements can change what constitutes a "best" model.
- Future forecasters may be interested in the development of ensemble forecasts that weight based on recent IAS performance, similarly to how the COVIDhub-trained-ensemble weights by the inverse of MWIS.
- Qualitative assessments via an allocation score-like framework, in which forecasts' allocations are visualized by location, may be useful for identifying "weak points": locations to which forecasts frequently under/overallocate.

Questions?

